



**UNIVERSITY OF
PERPETUAL HELP
SYSTEM DALTA**



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CogniCore: An Adaptive Cognitive Training Game with AI-Driven Skill Assessment and Analytics

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CHAPTER 1

THE PROBLEM AND REVIEW OF LITERATURE

1.0 Introduction

This chapter presents the background of the study, the identified problem and research gap, the proposed solution, project context, objectives of the study, scope and limitations, significance of the study, and the review of related literature and studies. It provides the foundation of the research by discussing the need for an adaptive cognitive training and assessment platform that utilizes modern technologies such as Artificial Intelligence, Machine Learning, and serious games.

Cognitive assessment tools have been developed due to technological advancements and have provided more effective approaches compared to traditional methods. Although traditional cognitive assessment methods remain widely used, they have several limitations related to test anxiety, user motivation, and behavioral assessment. As a result, researchers have explored game-based cognitive assessment approaches to improve the assessment process and enhance user engagement.

Game-based assessments have shown significant potential in improving user motivation and behavioral analysis. Furthermore, technologies such as Serious Games, Artificial Intelligence (AI), Machine Learning (ML), and Dynamic Difficulty Adjustment (DDA) have demonstrated their capability to provide more personalized and effective cognitive evaluation experiences. However, existing cognitive assessment tools primarily focus on cognitive exercises and performance scores while giving limited attention to cognitive profiling and behavioral analytics.



This study aims to address these limitations through the development of **CogniCore: An Adaptive Cognitive Training Game with AI-Driven Skill Assessment and Analytics**. The proposed system integrates game-based assessment, serious game mechanics, Dynamic Difficulty Adjustment, Machine Learning-based cognitive archetype classification, and behavioral analytics to evaluate users' cognitive abilities through gameplay interactions. By combining these technologies within a single platform, the study seeks to provide a more engaging, adaptive, and data-driven approach to cognitive assessment and training.

1.1 Problem and Its Background

1.1.1 Background of the Study

The development and growth of digital technologies have immensely impacted how people acquire education, undergo training, and even evaluate themselves on cognitive skills. Cognitive assessment is considered a crucial tool for evaluation of cognitive functions such as memory, concentration, reasoning, decision-making, problem-solving, and information processing ability. All these cognitive skills are vital in educational achievement, professional work, and even in regular activities. With the constant progress of technology, specialists have been researching new ways to assess and train cognitive skills effectively and efficiently. Historically, cognitive assessments were being done using standardized tests, questionnaires, and other psychological evaluations performed by experts. Although these approaches remain popular, they are associated with certain problems. Test anxiety, stress, boredom, and lack of motivation can be observed when undergoing assessments, thus adversely affecting the quality of results. Moreover, the traditional assessment tools remain static and yield no more information than numbers and letters, without providing any details about cognitive behavior and skills.



According to Berg (2021), game-based cognitive assessment tools improve user engagement and provide a more enjoyable assessment experience compared to traditional testing methods. The study revealed that integrating game mechanics into assessment activities encourages active participation while reducing the stress commonly associated with conventional cognitive evaluations. Similarly, according to Bhargava, Kottapalli, and Baths (2024), virtual reality and game-based cognitive assessments can effectively evaluate cognitive performance while maintaining user immersion and motivation. Their findings suggest that digital game environments have significant potential as alternative tools for cognitive assessment.

The increasing popularity of serious games has opened new opportunities for cognitive training and behavioral evaluation. Serious games are digital games designed for purposes beyond entertainment, including education, training, healthcare, and assessment. According to Vasconcelos et al. (2024), game-based cognitive assessments improve user motivation, usability, and participation while reducing stress during evaluation activities. Likewise, according to Wang et al. (2022), adaptive and enjoyable gameplay encourages long-term engagement and consistent cognitive data collection. These findings highlight the potential of serious games as effective tools for gathering cognitive and behavioral information in a more natural and engaging environment.

Some examples of cognitive training and brain-training apps include Lumosity (Lumos Labs, Inc., 2025), Peak (Peak Labs, 2025), Elevate (Elevate Labs, 2025), NeuroNation (Synaptikon GmbH, 2025), and CogniFit (CogniFit Inc., 2025). These tools include exercises that aim at enhancing memory, attention, processing speed, reasoning skills, and cognitive problem solving. In addition, performance tracking and personalized training are included to help develop cognitive abilities further. Although the applications mentioned above work effectively and deliver efficient cognitive exercises, they still lack a deeper analysis and focus on cognitive training and scores. Furthermore, they usually do not provide any useful



behavioral analytics and do not explain how users' performances can be explained through cognitive aspects of their actions and decision making.

Recently, due to the progress in the field of artificial intelligence (AI) and machine learning (ML), there have emerged new possibilities for conducting cognitive assessments using intelligent approaches. As Ahmad et al. (2023) report, machine learning approaches allow identifying meaningful behavioral patterns while assessing cognitive abilities from the information received during gameplays. At the same time, according to Tolks, Schmidt, and Kuhn (2024), artificial intelligence technology allows increasing personalization and adaptability as well as analyzing behavioral data within serious games.

In addition to personalized learning environments, game-based systems now use adaptive technologies, including Dynamic Difficulty Adjustment (DDA). Moon and Seo (2020) indicate that DDA contributes to user engagement since it involves automatic adjustments of game challenges depending on users' performance. In their turn, Chiotaki, Pouloupoulos, and Karpouzis (2023) point out that adaptation can contribute to the positive results of the application since it makes the system learn from the individual users and customize game sessions accordingly, which can improve users' satisfaction and learning efficiency. With the help of adaptive difficulty mechanisms, users will be able to stay engaged without experiencing frustration or boredom.

Despite the increased popularity of such applications, there are still many difficulties preventing the use of efficient cognitive assessment tools. As Berg (2021) explains, cognitive assessments must include not only performance measurements but also maintenance of user engagement to increase the reliability of their outcomes. Unfortunately, some traditional assessments still cause anxiety and demotivation of their participants. Moreover, as pointed out by Moon and Seo (2020), adaptive mechanisms that allow adjusting games to the user performance are needed for engagement. At this point, most cognitive training platforms lack Dynamic Difficulty Adjustment, which leads to boredom and frustration.



Additionally, Ahmad et al. (2023) argue that ML technology has the ability to detect patterns and behavior from user activities. Although this is possible, current cognitive training apps are only interested in giving numerical scores instead of behavioral analysis and cognitive profiling. Additionally, Tolks, Schmidt, and Kuhn (2024) observe that although artificial intelligence can help personalize the cognitive training process and analyze behavior, current apps lack the use of AI in providing cognitive insight. Thus, the user does not get enough feedback on their cognitive strength, weaknesses, and decision-making styles.

1.1.2 Research Gap

The reviewed studies demonstrate that game-based cognitive assessments are effective in evaluating different cognitive abilities through interactive and engaging activities. Studies by Hadjiaros et al. (2021), Knobel et al. (2021), and Müller et al. (2024) showed that reaction-time tasks, visual search activities, and Stroop-based assessments can effectively measure processing speed, attention, cognitive control, and inhibitory functions. Similarly, Bhargava et al. (2024) and Vasconcelos et al. (2024) emphasized the effectiveness of game-based assessments in evaluating memory, attention, and spatial retention. Studies by Pedersen et al. (2020), Wang et al. (2022), Schubert et al. (2023), and Nogueira et al. (2021) further demonstrated that game-based environments can assess logical reasoning, problem-solving, pattern recognition, and strategic planning skills. Moreover, Chiotaki et al. (2023) highlighted the benefits of adaptive game-based systems in promoting cognitive flexibility, strategic thinking, and user engagement.

Despite these advancements, most existing studies focus on evaluating specific cognitive domains independently. Reaction-time tasks primarily assess attention and processing speed, memory-based games focus on memory retention, while reasoning and strategy games evaluate logical thinking and decision-making separately. Few studies combine these cognitive domains within a single platform capable of providing a comprehensive assessment of memory, attention, reasoning, reflexes, and executive functioning.



Furthermore, most game-based cognitive assessment systems generate performance scores but provide limited behavioral analytics and cognitive profiling. While some studies explore adaptive gameplay and personalized learning experiences, there remains a lack of systems that integrate Dynamic Difficulty Adjustment (DDA), Machine Learning-based cognitive classification, behavioral analytics, and cognitive archetype generation within one web-based environment.

Therefore, there is a need for a unified cognitive assessment platform that combines multiple cognitive domains, adaptive gameplay, Artificial Intelligence, Machine Learning, and behavioral analytics to provide deeper cognitive insights beyond traditional scoring methods. This gap serves as the foundation for the development of **CogniCore: An Adaptive Cognitive Training Game with AI-Driven Skill Assessment and Analytics**, which aims to deliver a comprehensive, engaging, and data-driven approach to cognitive assessment and training.

1.1.3 Proposed Solution

This study attempts to bridge this gap by developing an innovative and effective system known as CogniCore: An Adaptive Cognitive Training Game With AI-Driven Skill Assessment and Analytics. This will include various cognitive tests assessing different aspects of cognitive ability, namely Memory, Attention, Reasoning, and Executive Functioning, on a single web-based platform. Furthermore, the proposed system includes features such as DDA, cognitive archetypes through ML, behavioral analysis, and performance visualization.

1.2 Project Context

CogniCore is a web application which serves as a platform where the users can assess and develop their cognitive abilities. CogniCore uses four different mini games based on Spatial



Visual memory, Logical Mathematical reasoning, Reflex & attentional focus, and executive strategy.

The system is available via web browser applications. Technologies used in this platform include React.js, Phaser.io, Python Flask, Scikit-learn, and MySQL. The system receives information about the users' performance from their interaction within the game and analyzes the information using Machine learning techniques, after which the cognitive archetypes and behavioral analysis reports are generated.

1.3 Objectives of the Study

1.3.1 General Objective

To develop an Adaptive Cognitive Training Game with AI-driven Skill Evaluation and Behavioral Analytics, the purpose is to build a web-based game platform which will make use of AI technology, machine learning, and dynamic difficulty adjustment for cognitive skill assessment and behavior analytics.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. To develop four cognitive games which will test for:
 - Spatial/Visual Memory
 - Logical/Mathematical Reasoning
 - Reflexes and Concentration
 - Strategic Thinking
2. To implement a DDA (Dynamic Difficulty Adjustment) system that automatically adjusts the level of difficulty in relation to the level of the players' proficiency.



3. To Create a game data tracking system which tracks reaction time, accuracy, decision-making patterns, and cognitive performance.
4. To implement a machine learning model for classification called Random Forest Classifier from Scikit-Learn using a Python Flask backend to classify users into different Cognitive Archetypes.
5. To Create a web application to track cognitive performance using interactive analytics dashboard created using React.js & Chart.js.
6. Provide users with personalized cognitive insights and performance reports through AI-driven behavioral analytics.
7. 7.Evaluate the developed system using the **Game Experience Questionnaire (GEQ)** in terms of:
 - Competence
 - Sensory and Imaginative Immersion
 - Flow
 - Tension and Annoyance
 - Challenge
 - Positive Affect
 - Negative Affect
8. Evaluate the effectiveness of application using pretest and posttest

1.4 Scope and Limitation

1.4.1 Scope of the Study

The research is aimed at the development and design of a cognitive assessment and training platform titled CogniCore: An Adaptive Cognitive Training Game with AI-Driven Skill Assessment and Analytics. This system consists of four interactive cognitive mini-games that are intended for testing Spatial-Visual Memory, Logical-Mathematical Reasoning, Reflexes



& Focus of Attention, and Executive Strategy. The system features user registration and authentication, real-time Dynamic Difficulty Adjustment (DDA), cognitive classification utilizing Machine Learning techniques, behavioral analysis, performance monitoring, and data visualization by means of an interactive dashboard. React.js is used for developing the user interface of the system, Phaser.io is chosen as the game engine, Python Flask framework and Scikit-learn library for Machine Learning, and MySQL as the database management system.

1.4.2 Limitations of the Study

The study only acts as a non-medical assessment tool and training platform for cognitive performance and is not aimed at diagnosing any neurological condition, psychological issues, mental disorder, or any learning disability. The study only analyzes the cognitive functions of an individual using the gameplay and behavioral analysis of the cognitive mini-games created for assessing the four cognitive domains. In addition, the study is limited to the creation of only four cognitive domains, and any other domains will fall out of the ambit of the study. The study demands an active internet connection because of its dependence on real-time data processing, machine learning, and dashboard generation. Moreover, the current system is tailored to accommodate only English language users and can be accessed only using modern web browsers.

1.5 Significance of the Study

Students and General Users

The system provides a stress-free, accessible, and engaging platform for tracking cognitive growth and behavioral patterns without the high costs associated with professional psychological profiling.



Educators and Institutions

The study provides institutions with interactive behavioral analytics and cognitive performance reports beyond traditional numerical scoring systems.

Game Developers

The project demonstrates how Artificial Intelligence and serious games can be integrated for cognitive data gathering, adaptive learning, and behavioral analytics beyond entertainment purposes.

Future Researchers

The study may serve as a technical and academic reference for future studies involving Artificial Intelligence, Machine Learning, Adaptive Gameplay Systems, Serious Games, Cognitive Analytics, and Behavioral Data Mining.

1.6 Review of Related Literature and Studies (RRL/RRS)

Cognitive Assessment and Serious Games

Cognitive assessments can be really stressful. Make people feel anxious. They are also not always easy to access. Berg did a study in 2021. Created a game that you can play online to test how well your brain is working. The study found out that playing games to test your brain is more fun. People get less tired than when they take traditional tests.

Some other people, Bhargava, Kottapalli and Baths did a study in 2024. They compared playing games on your phone or using reality to traditional tests. What they found out was that playing games is a way to test how well your brain is working and it is also more fun.



Vasconcelos and some other people looked at how easy it's to use games to test your brain. They found out that games that are meant to be fun and educational really do make people want to play them and get involved in activities that're good for their brain.

Wang and some other people did a study in 2022. They looked at how people can focus when they are playing games that test their brain. What they found out was that when the games are fun and get harder as you get better people want to keep playing and they stay interested for a time. Cognitive assessments, like these games are a way to test cognitive functions and they can be really helpful. Game-based cognitive assessments are a way to make cognitive assessments more fun and accessible.

Synthesis

The reviewed studies demonstrate that serious games provide a more engaging and less stressful alternative to traditional cognitive assessments. These findings support the development of CogniCore as a game-based cognitive assessment platform that promotes user engagement while gathering meaningful cognitive data.

Dynamic Difficulty Adjustment and Adaptive Gameplay

Adaptive gameplay systems are used in games to keep players engaged and to balance challenge levels.

Chiotaki, Pouloupoulos and Karpouzis (2023) found that adaptive systems improve engagement, personalization and learning outcomes by adjusting gameplay based on player performance. This is achieved by making games more adaptive to players.

Moon and Seo (2020) proposed a model that adjusts game difficulty quickly based on player performance. They found that such adaptive systems improve immersion and motivation in games. Adaptive difficulty systems help to make games more engaging.



Nebel et al. (2020) Highlighted that a balance between challenge and skill level is critical for a gaming experience. The psychological Flow State during gameplay is maintained when the game is not too easy or too hard.

Seyderhelm and Blackmore (2021) emphasized the need for mechanisms in serious game environments. These mechanisms help to keep players engaged and motivated.

Van der Linden et al. (2021) Concluded that adapting game mechanics based on player behavior has an effect on immersion, engagement and effectiveness. Adaptive gameplay systems help to achieve this.

Synthesis

The reviewed studies show that adaptive gameplay systems improve engagement reduce frustration and keep players motivated. These findings support the use of Dynamic Difficulty Adjustment in CogniCore. Adaptive gameplay systems are useful, for games. They make games more engaging and fun to play.

Machine Learning and AI-Driven Analytics

Artificial Intelligence and Machine Learning technologies are increasingly used in serious games for personalization, analytics, and behavioral evaluation.

Ahmad et al. (2023) proposed a game-based intelligent system for evaluating clusters of cognitive abilities through Machine Learning techniques.

Tolks, Schmidt, and Kuhn (2024) found that AI improves personalization, adaptive gameplay, analytics generation, and user engagement.

Wali et al. (2025) revealed that AI algorithms are widely used for adaptive gameplay, real-time data processing, and behavioral assessment.



Zapušek et al. (2023) emphasized that Machine Learning models improve adaptive systems by analyzing user behavior patterns and adjusting system responses accordingly.

Synthesis

The reviewed studies indicate that Artificial Intelligence and Machine Learning enhance personalization, analytics, and cognitive performance evaluation. These findings support the integration of Random Forest Classification and AI-driven behavioral analytics within CogniCore.

Web-Based Technologies and Interactive Platforms

Modern serious games increasingly utilize web-based technologies to improve accessibility and scalability.

A study published in *Frontiers in Psychology* (2021) demonstrated that gameplay analytics and real-time behavioral tracking can effectively evaluate user cognitive performance.

Gao et al. (2023) highlighted the importance of interactive dashboards, data visualization, and real-time analytics in improving user understanding and system usability.

The *Journal of Medical Internet Research* (2023) emphasized that web-based serious games improve accessibility, engagement, and continuous participation compared to traditional systems.

Wang et al. (2021) found that serious games positively contribute to cognitive stimulation and user engagement while maintaining accessibility and interactivity.



Synthesis

The reviewed studies support the use of web-based technologies such as React.js, Phaser.io, Python Flask, Chart.js, and MySQL in developing CogniCore as an accessible and interactive AI-powered cognitive assessment platform.

Working memory is an important cognitive skill that helps people remember and use information for a short period of time. It is used when solving problems, following instructions, making decisions, and learning new things. Haque, Samandra, and Alizadeh Mansouri conducted a study in 2021 about working memory. They explained that working memory helps people temporarily store and process information while performing different tasks.

Working Memory

Working memory is an important cognitive skill that helps people remember and use information for a short period of time. It is used when solving problems, following instructions, making decisions, and learning new things. Haque, Samandra, and Alizadeh Mansouri conducted a study in 2021 about working memory. They explained that working memory helps people temporarily store and process information while performing different tasks.

Working memory is important in everyday activities because people use it when remembering numbers, following directions, and completing tasks that require concentration. Games that involve remembering patterns, matching objects, and recalling sequences can help assess and improve working memory performance.



Synthesis

The reviewed literature shows that working memory plays a significant role in learning, problem-solving, and decision-making. These findings support the development of the Memory Match module in CogniCore as a game-based approach for assessing and improving working memory skills.

Spatial Memory

Spatial memory is the ability to remember locations, directions, and the arrangement of objects in an environment. It helps people navigate places, recognize patterns, and recall visual information. Jonson and colleagues conducted a study in 2021 and found that digital and virtual environments can effectively assess spatial memory and visuospatial skills.

Spatial memory is important for navigation and environmental awareness. It allows individuals to remember where objects are located and how to move from one place to another. Interactive games that require players to remember positions and patterns can help measure spatial memory performance.

Synthesis

The reviewed literature indicates that spatial memory is essential for navigation, visual learning, and environmental awareness. These findings support the inclusion of the Matrix Recall module in CogniCore for assessing spatial retention and recall abilities.

Logical Reasoning

Logical reasoning is the ability to analyze information, identify patterns, and make sound decisions based on evidence. It plays an important role in problem-solving, critical thinking,



and decision-making. Weng conducted a study in 2022 and found that puzzle-based games can improve logical thinking and analytical skills.

Logical reasoning helps individuals solve problems by applying rules and recognizing relationships between different pieces of information. Educational games that involve puzzles and pattern recognition can effectively assess reasoning abilities while maintaining user engagement.

Synthesis

The reviewed literature demonstrates that logical reasoning is essential for analytical thinking and effective problem-solving. These findings support the development of the logical reasoning modules in CogniCore to evaluate users' reasoning and decision-making skills.

Executive Strategy and Cognitive Flexibility

Executive strategy refers to the ability to plan, organize, make decisions, and solve complex problems. Cognitive flexibility is the ability to adapt to new situations and change strategies when needed. Chiotaki, Pouloupoulos, and Karpouzis conducted a study in 2023 and found that adaptive game-based learning systems can improve strategic thinking and cognitive flexibility through adaptive challenges.

These skills are important because they help individuals adjust to new circumstances, think critically, and make effective decisions. Games that require planning, route selection, and adapting to changing rules can help assess executive functioning and strategic thinking.

Synthesis



The reviewed literature shows that executive strategy and cognitive flexibility are important cognitive skills that support planning, adaptability, and decision-making. These findings support the inclusion of executive strategy modules in CogniCore.

Attention and Processing Speed

Attention and processing speed refer to the ability to focus on relevant information and respond quickly to stimuli. These cognitive skills are important for learning, decision-making, and task performance. Hadjiaros and colleagues conducted a study in 2021 and found that game-based cognitive assessments can effectively measure reaction time, attentional control, and processing speed.

Individuals with stronger attention and processing speed can respond more accurately and quickly to changing situations. Interactive cognitive games that require rapid responses and visual attention provide useful measures of these cognitive abilities.

Synthesis

The reviewed literature indicates that attention and processing speed are essential cognitive skills for effective learning and decision-making. These findings support the development of the attention and reflex-based modules in CogniCore.

Related Systems

Speed Tap (Processing Speed and Reaction Time)

According to Hadjiaros et al. (2021), game-based cognitive assessments that incorporate reaction-time tasks can effectively measure visuospatial attention and response speed. Their study validated gamified cognitive tasks and found that participants exhibited performance patterns similar to traditional cognitive assessments. This suggests that reaction-based games



can serve as reliable tools for measuring cognitive processing speed and motor responses. The findings support the implementation of the **Speed Tap** module in CogniCore, which measures users' reaction time, response accuracy, and processing speed through time-constrained gameplay activities.

Focus Finder (Selective Attention and Visual Search)

Knobel et al. (2021) developed a visual search task within an immersive game environment to assess attention and target identification abilities. The researchers found that adaptive visual search activities effectively evaluate selective attention and visual scanning performance while maintaining user engagement. The study demonstrated that search-based cognitive tasks can be enhanced through game mechanics and adaptive difficulty systems. This supports the development of the **Focus Finder** module, which requires users to identify target objects among distractors and evaluates visual attention and focus.

Visual Search and Attention Control

Park and Zhang (2021) investigated visual search performance under conditions involving distractor interference. Their findings revealed that visual attention tasks are effective in measuring selective attention, cognitive control, and distractor resistance. The study emphasized that visual search tasks can provide valuable indicators of attentional performance and information processing efficiency. These findings support the use of visual search mechanics in CogniCore's **Focus Finder** module to evaluate attentional focus and concentration.

Stroop Shift (Cognitive Control and Inhibitory Attention)

Müller et al. (2024) examined different versions of the Stroop task and concluded that Stroop-based activities remain effective tools for measuring selective attention, cognitive flexibility, and inhibitory control. The study highlighted the ability of Stroop tasks to



evaluate how individuals manage conflicting information and suppress automatic responses. These findings support the integration of the **Stroop Shift** module in CogniCore, which challenges users to identify colors while ignoring conflicting textual information.

Memory Match (Working Memory)

Bhargava, Y., Kottapalli, A., & Baths, V. (2024). Validation and comparison of virtual reality and 3D mobile games for cognitive assessment against ACE-III in young participants. *Scientific Reports*, 14. This study found that game-based cognitive assessments effectively evaluate memory, attention, and executive functions while maintaining user engagement and immersion. The findings support the use of game-based memory tasks such as Memory Match for assessing working memory performance.

Matrix Recall (Spatial Memory)

Vasconcelos et al. (2024) reported that game-based cognitive assessment systems effectively measure spatial memory, attention, and navigation skills while improving user engagement compared to traditional assessment methods. Their systematic review concluded that interactive cognitive games provide reliable measures of cognitive performance and memory retention. This supports the implementation of the Matrix Recall module in CogniCore.

Logic Link (Inductive Logic)

According to Pedersen et al. (2020), game-based cognitive assessments can effectively measure multiple cognitive abilities, including reasoning, problem-solving, memory, and decision-making through interactive gameplay. Their validation study involving more than 10,000 participants demonstrated that in-game behavioral data can be used to predict cognitive abilities with reliability comparable to traditional assessments. These findings support the implementation of the Logic Link module in CogniCore, which evaluates logical reasoning and pattern recognition through game-based interactions.



Equation Balance (Deductive Logic)

Mathematical reasoning tasks evaluate analytical thinking, procedural knowledge, and numerical problem-solving abilities. According to Wang et al. (2022), game-based arithmetic challenges can effectively measure cognitive processing speed and deductive reasoning performance. These findings support the implementation of the **Equation Balance** module.

Sequence Decoder (Pattern Recognition)

Pattern recognition and sequence analysis are fundamental components of inductive reasoning. According to Schubert et al. (2023), sequence-based cognitive tasks effectively evaluate abstract reasoning and rule discovery abilities. Their findings indicate that identifying hidden patterns improves assessment accuracy for analytical thinking skills.

Route Optimizer (Combinatorial Logic)

Route planning and optimization tasks assess strategic reasoning, decision-making, and resource allocation abilities. According to Nogueira et al. (2021), pathfinding and route optimization games can effectively measure executive planning and combinatorial problem-solving skills. These findings support the inclusion of the **Route Optimizer** module.

Maze Escape and Mental Flex (Strategic Problem Solving)

According to Chiotaki, Pouloupoulos, and Karpouzis (2023), adaptive game-based systems can effectively enhance problem-solving skills, strategic thinking, and cognitive flexibility by adjusting challenges according to user performance. Their systematic review found that adaptive gameplay improves engagement and supports the development of higher-order cognitive abilities. These findings support the implementation of the **Maze Escape** and



Mental Flex modules in CogniCore, which evaluate users' planning abilities, decision-making skills, and adaptability to changing conditions.

Synthesis

The reviewed related systems and studies demonstrate the growing effectiveness of game-based cognitive assessments in evaluating various cognitive domains, including attention, memory, reasoning, problem-solving, cognitive flexibility, and executive functioning. Studies on reaction-time tasks such as Speed Tap revealed that game-based assessments can effectively measure processing speed, response accuracy, and visuospatial attention. Similarly, visual search and attention-control studies support the use of Focus Finder and Stroop Shift in assessing selective attention, inhibitory control, and cognitive flexibility through engaging gameplay activities.

Research on memory-based cognitive games further established the effectiveness of game mechanics in evaluating working memory and spatial retention. The findings of Bhargava et al. (2024) and Vasconcelos et al. (2024) suggest that interactive cognitive games provide reliable measures of memory performance while maintaining user engagement. These studies support the implementation of Memory Match and Matrix Recall as cognitive assessment modules within the proposed system.

Studies involving logical reasoning, pattern recognition, mathematical problem-solving, and strategic decision-making also highlight the potential of game-based environments for assessing higher-order cognitive abilities. The works of Pedersen et al. (2020), Wang et al. (2022), Schubert et al. (2023), and Nogueira et al. (2021) demonstrate that gameplay data can be utilized to evaluate reasoning, analytical thinking, pattern recognition, and strategic planning with results comparable to traditional cognitive assessments. These findings provide a strong foundation for the Logic Link, Equation Balance, Sequence Decoder, and Route Optimizer modules of CogniCore.



Furthermore, research on adaptive game-based learning conducted by Chiotaki, Pouloupoulos, and Karpouzis (2023) emphasizes the importance of adaptive gameplay in promoting cognitive flexibility, problem-solving, and user engagement. Their findings support the integration of Dynamic Difficulty Adjustment (DDA) within the Maze Escape and Mental Flex modules, enabling the system to personalize challenges according to user performance.

Overall, the reviewed studies confirm that serious games can serve as effective tools for cognitive assessment by providing engaging and data-rich environments for measuring cognitive abilities. These findings support the development of CogniCore as a web-based cognitive training and assessment platform that combines memory, attention, reasoning, executive functioning, adaptive gameplay, and behavioral analytics into a unified system.

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Chapter 2:

METHODOLOGY

This chapter presents the systematic approach and technical framework utilized in the development of CogniCore. It details the research design, the selection of participants, and the specialized tools used to bridge game mechanics with machine learning. Furthermore, this section outlines the software development life cycle, the architectural diagrams defining data movement, and the rigorous testing procedures required to validate the system's effectiveness as a cognitive assessment tool.

2.1 Purpose and Description of the Study

Purpose

This project aims at developing a cognitive training game called CogniCore that will try to overcome the problems of traditional methods of evaluation by making use of an adaptive game form. By cloaking the test in an enjoyable format, it seeks to avoid the problem of "test anxiety" that leads to distortion of performance in terms of working memory and accuracy.

Description

CogniCore is a serious game platform on the internet that makes use of artificial intelligence to analyze player performance. Some of the characteristics of CogniCore include four 2D interactive games designed using Phaser.io technology, a real-time Dynamic Difficulty Adjustment (DDA) engine that automatically adjusts the complexity of the game depending on players' performance, and machine learning software using Python. The intended users of CogniCore are learners and other people who want to track their own performance.



2.1.2 Operational Definition of Terms

- In order to achieve absolute clarity on valuing the systems and conducting software audits, the following is how the various technical and structural components of the CogniCore framework are contextually operationalized:
- Artificial Intelligence (AI): The overall software engineering framework that works on the Python server machine to interpret unstructured participant input data, tune the system task complexity, and provide prediction based on complicated streams of data without hard-coded commands.
- Machine Learning (ML): The specific statistical system which operates within the /api/dda endpoint and uses structural estimation techniques to learn classification schemes for performance features from gaming actions.
- Random Forest Classifier: Machine learning model that is set up with a list of 50 decision trees having a maximum depth constraint of 5, used in calculating confidence scores for human archetype classification based on past moving average values for hit rates, reaction time, and error latency.
- Behavioral Analytics: The constant algorithmic surveillance of micro-state interactions in the frontend canvas layer involving non-target cursor delay actions and panicked input clusters aimed at capturing player psychology patterns beyond the typical score metrics.
- Cognitive Radar Chart: Interactive multi-axis vector visualization canvas drawn at the frontend dashboard level via React.js and Chart.js, specifically to visualize real-time levels of proficiency within all four primary structural academic domains.
- Hesitation Latency (hesitation_ms): Absolute duration time measured in milliseconds from the time the visual object appears until the time when the subject first manipulates the pointer on the screen.
- Spam Click Count (spam_click_count): This is an additional behavioral tracking metric that increments automatically whenever there is a speed of clicks above 5 clicks per second in the dead zones on a blank canvas.



- **Cognitive Domain Inheritance:** The state-preserving mechanism through the implementation of the /api/start-session endpoint where an individual's assigned challenge level is automatically passed on to sibling services belonging to the same parent subject category, thus setting the initial floor to the maximum ceiling of the user's existing baseline.

2.2 Research Design

The researchers will utilize the Developmental-Descriptive Research Design. The developmental aspect covers the engineering of the platform, including the design of the Phaser.io games and Python AI backend. The descriptive aspect focuses on the evaluation of the system using the ISO 25010 Software Quality Model and the Game Experience Questionnaire (GEQ).

2.3 Participants or Respondents

The researchers identified three distinct groups of participants to ensure the technical and psychological validity of the system.

A. Focus Group (20 Participants)

A dedicated focus group consisting of twenty (20) individuals will be selected. This group is required to undergo the full training program within the game and participate in the Pre-test and Post-test sessions. Their data is essential for the researchers to calibrate the AI's learning curves and verify the effectiveness of the training modules.

B. Evaluation Group (100 Respondents)

To measure the system's usability and engagement, the researchers will recruit 100 respondents. This group represents a diverse demographic, including both regular students



and experienced gamers. They will provide feedback via the ISO 25010 and GEQ surveys to ensure the platform is accessible and engaging for a wide variety of users.

C. Technical Experts (10 IT Professionals)

Ten (10) IT experts will serve as technical validators. They will evaluate the source code, the efficiency of the Random Forest model, and the security of the database management system. Their professional feedback will ensure that the system meets industry standards for software engineering.

B. Technical and Subject Matter Experts (The Validators)

To validate the system's architecture and the scientific basis of the game modules, a panel of experts will be consulted.

- **Sample Size: Three (3) to Five (5) experts.**
- **Expert Criteria:**
 1. **IT/Game Development Experts:** Specialists or professors who have knowledge of software engineering, web development, or artificial intelligence. They will analyze the efficiency of the code along with integration of the artificial intelligence engine developed in the Python language.
 2. **Subject Matter Experts (SMEs):** Qualified Guidance Counselors or Psychology Instructors. These people will ascertain that the mini-games meet the requirement of "content validity," meaning that the games will measure memory, logic, reflexes, and strategy in an effective manner suitable to the age group.

- **Role in the Study:** These specialists will evaluate the system according to ISO 25010 quality standards to ensure that the generated "cognitive profiles" are logically sound in respect of the users' real performance in games.

2.4 Project Development Methodology

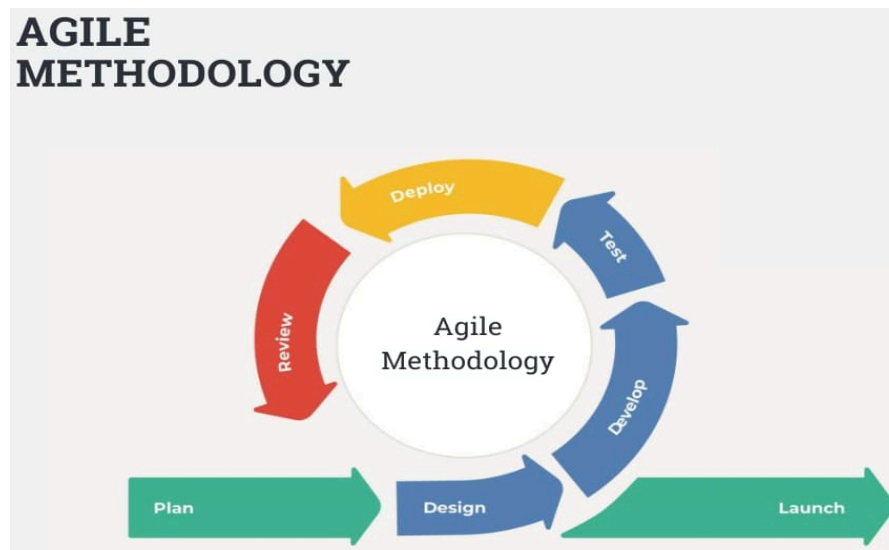


Figure 1: The Agile Development Cycle

The researchers selected the Agile Development Methodology to manage the technical complexities of integrating game mechanics with machine learning. This methodology is characterized by its iterative nature, allowing for continuous refinement based on real-time testing. Below are the official steps of the Agile process and their specific application in this study:

1. **Requirement Analysis:** This initial phase involves gathering all functional and non-functional requirements to define the project's scope. For CogniCore, the proponents focused on identifying the four cognitive domains (Memory, Logic, Reflex, and Strategy) and selecting the appropriate tech stack, including Phaser.io and Python, to ensure the system meets the research objectives.



2. **Design:** In this stage, the architectural framework and user interface are planned. The team utilized this phase to create UI mockups in Figma and draft the technical diagrams, such as the Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD). This ensures that the data flow between the game engine and the AI backend is logically mapped before coding begins.
3. **Development:** This is the construction phase where the actual programming occurs. The researchers will develop the game in "Sprints," focusing on one mini-game module at a time. This allows for the simultaneous coding of the Phaser.io frontend and the Python/Flask backend, ensuring that the Dynamic Difficulty Adjustment (DDA) scripts are functioning correctly within the game environment.
4. **Testing:** This phase involves identifying and fixing bugs through various testing levels. The proponents will conduct unit testing on each game module to verify scoring accuracy and AI classification precision. By testing early and often, the team can resolve technical issues before they affect the overall system performance.
5. **Implementation:** During this stage, the software is deployed to the target environment. The researchers will host the CogniCore platform on cloud-based services like Vercel and Render, making it accessible to the respondents via web browsers. This phase ensures that the system is stable and ready for the official data-gathering sessions.
6. **Review:** The final phase involves evaluating the system based on user feedback and expert validation. The team will conduct a Pilot Run with the respondents and consult with IT experts to review the system's quality based on ISO 25010 standards. The insights gained from this review will be used to perform the final polish of the documentation and the software



2.5 Requirements Specifications

Table 1: Tools, Technologies, or Platforms Used

Category	Technology Used
Frontend UI/UX	React.js, HTML5, CSS
Game Engine	Phaser.io (JavaScript)
Backend & AI Server	Python (Flask), Scikit-learn
Database	MySQL
IDE	Visual Studio Code

Tools, Technologies, or Platforms Used presents the software tools, frameworks, and technologies utilized in the development of the CogniCore system. The table identifies the technologies used for the frontend user interface, game development, backend processing,



artificial intelligence implementation, database management, and development environment. These tools work together to support the functionality, performance, and deployment of the web-based cognitive training game, enabling interactive gameplay, AI-driven cognitive assessment, data storage, and system maintenance.

2.5.1 - Functional Requirements

- **User Authentication:** Secure login and registration for tracking individual player history.
- **Real-Time Data Tracking:** The system must record micro-actions (clicks, speed, errors) during Phaser.io gameplay.
- **Procedural Generation:** The game must dynamically scale its difficulty variables (speed, visual noise) while a level is actively being played.
- **Report Generation:** The system must process the completed game data through the AI model and render a visual Radar Chart on the user dashboard.

2.5.2 - Software Requirements

The following software and tools are essential for the development and operation of the CogniCore platform:



Table 2: Software Requirements

Category	Minimum	Recommended	Development
Operating System	Windows 10	Windows 10 or higher	Windows 11 Pro
Browsers	Chrome, Edge	Chrome, Opera GX	Chrome (Latest)
Third-party Apps	Web Browser	Python 3.9 Runtime	VS Code, Git, Figma
Programming Language	HTML5, CSS	JavaScript (ES6+)	Python (Flask), JS

Software Requirements presents the software specifications required for the operation and development of the CogniCore system. The table identifies the minimum, recommended, and development software requirements, including the operating system, web browsers, third-party applications, and programming languages. These software components provide the necessary environment for running the web-based cognitive training game, supporting system development, and ensuring compatibility, functionality, and optimal performance across different platforms.

2.5.3 - Hardware Requirements

The hardware specifications are divided into developer requirements for building the AI and game logic, and user requirements for playing the game.



Table 3: Hardware Requirements

Category	Minimum	Recommended	Development
Processor	Intel Core i3	Intel Core i5	Intel Core i7
Memory	8GB of RAM	8GB of RAM	16GB of RAM
GPU	Intel HD Graphics	GTX 1050 or higher	RTX 3060 or higher
Storage	1GB available	5GB available	50GB available (SSD)

Hardware Requirements presents the minimum, recommended, and development hardware specifications needed to run and develop the CogniCore system. The table identifies the required processor, memory, graphics capability, and storage capacity for different usage levels. These specifications ensure that the web-based cognitive training game operates smoothly for end users while providing sufficient resources for system development, testing, and AI model implementation. The hardware requirements serve as a guide for achieving optimal system performance and user experience.

2.6 Data Gathering Procedures

To ensure the effectiveness and technical accuracy of CogniCore, the researchers will employ a multi-method approach to collect data during the different phases of development.

Survey

- Participants will fill out the structured survey questionnaires following one entire session of playing the game by thirty (30) students.
- The questionnaire will follow the framework of the ISO 25010 software quality model, specifically Usability, Functional Suitability, and Performance Efficiency.



- Feedback will be gauged using the Likert scale (i.e., from 1 to 5) to assess the level of engagement with the game and the reliability of the artificial intelligence generated profiles.

Interview

- Semi-structured interviews with the Subject Matter Experts (SMEs) will be conducted, especially licensed psychologists or guidance counselors.
- The purpose will be to obtain expert views concerning whether the mini-game aspects (memory, logic, reflex, and strategy) have any logical relationship with cognitive assessment criteria in reality.
- IT Professionals will also be interviewed to evaluate the efficiency of machine learning and the effectiveness of the MySQL database management.

Document Reviews

- The team will review existing literature and documents related to Serious Games and Machine Learning algorithms (Random Forest Classifiers).
- They will review documents related to standardized cognitive assessment tests, such as Stroop Test and Working Memory tasks, in order to use them as inspiration for the development of the games.
- Phaser.io engine and Scikit-learn documents will be reviewed for optimal game design and proper scaling of the AI's data.

Observation

- As part of the Pilot Run, the team will conduct "direct observation" during gameplay.
- The team will make observations on user activity, specifically any points where there is user confusion, points where there is frustration, or when the users enter a "Flow State."
- The observations will help the team confirm whether the DDA is functioning properly according to the user's activity.



- Observation data will be used to manually verify if the **Dynamic Difficulty Adjustment** (DDA) is responding correctly to the player's visible performance.

Testing Scripts / Code

- **Unit Testing:** There will be specific scripts to check each component of the game separately, for example whether the player's "Micro-actions" (i.e. clicking and reaction time) are getting registered in the database or not.
- **AI Model Validation:** To validate the Random Forest model, "Test-Train Split" techniques will be used in Python. The idea is to input some "known" data into the artificial intelligence to check whether it classifies the user into the right Cognitive Archetype.
- **Stress Testing:** Scripts will be written in order to simulate many people logging into the web application at once, in order to make sure that the Python Flask server and MySQL database are stable.

2.7 System Development Procedures

The system will be developed using Agile SDLC and will undergo iterative development and testing.

2.7.1 - Planning

The planning phase involves the strategic organization of the project's technical and creative requirements. The researchers focused on:

- **Module Selection:** Identifying the specific games for Memory, Logic, Reflex, and Strategy based on psychological benchmarks.
- **Architecture Design:** Planning the bridge between the Phaser.io frontend and the Python/Flask AI backend.

- **Risk Management:** Identifying potential "feature creep" and ensuring the AI model remains within a manageable scope for a web environment.

2.7.1.1 - Project Schedule: Gantt Chart

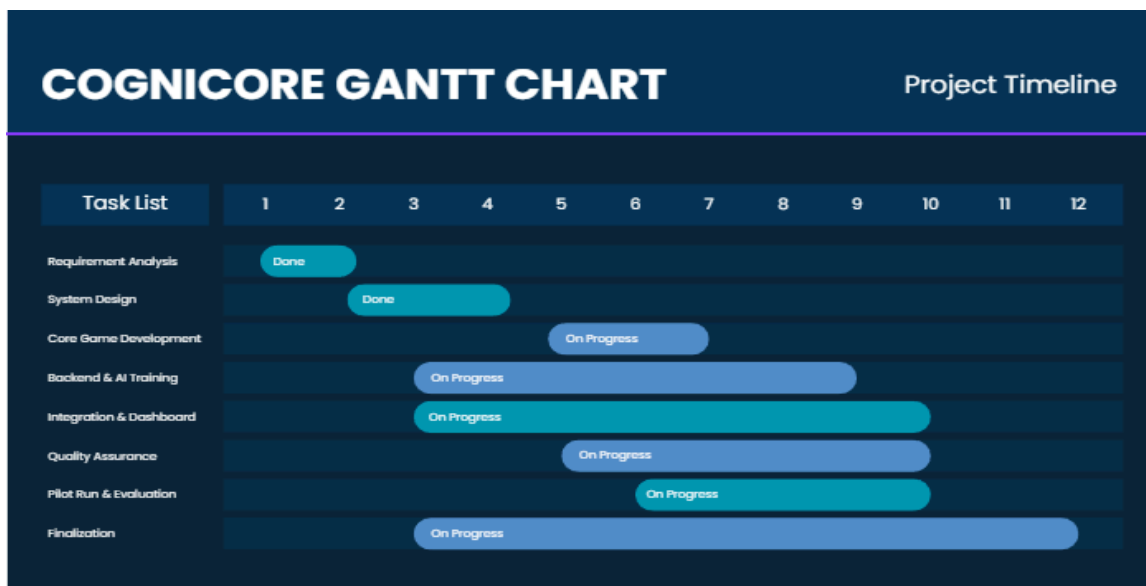


Figure 2: Timeline

1. **Requirement Analysis:** Our team will start by defining the four specific cognitive domains and setting up our collaborative Git repository to manage the source code (Weeks 1-2).
2. **System Design:** During this phase, the researchers will draft the standardized DFDs and ERDs while simultaneously creating interactive UI mockups in Figma (Weeks 3-4).
3. **Core Game Development:** This is the primary construction phase where our team programs the four interactive Phaser.io modules and the real-time DDA logic (Weeks 5-7).



4. **Backend & AI Training:** The team will focus on developing the Python/Flask API and training the Random Forest model to ensure accurate player categorization (Weeks 8-9).
5. **Integration & Dashboard:** We will connect the React.js analytics dashboard to the live MySQL database to allow for real-time visualization of gameplay data (Week 10).
6. **Quality Assurance:** In this period, the researchers will conduct rigorous unit tests and resolve any discovered gameplay bugs or data synchronization errors (Week 10).
7. **Pilot Run & Evaluation:** Our team will officially deploy the system to our 30 respondents and perform the necessary statistical analysis on the results (Week 11).
8. **Finalization:** To conclude the project, we will perform the final documentation polish and prepare all materials for the official Capstone defense (Week 12).

2.7.1.2 - Feasibility Study

To evaluate the financial viability of CogniCore, the researchers conducted a comprehensive budget analysis tailored to the project's lifecycle. This study distinguishes between the resources consumed during the building phase and those required for deployment. The costs are categorized into the Development Phase, which accounts for the technical labor and daily overhead, and the Implementation and Maintenance Phase, which covers the infrastructure needed to host the system for respondents and experts.

Table 4: Estimated Project Development and Operational Costs

Category	Specific Line Items	Estimated Cost (PHP)
I. Development Costs		
Labor (Simulated)	Technical effort for 3 researchers (600 combined man-hours @ ₱150/hr)	₱90,000.00
Connectivity	High-speed Fiber Internet	₱8,500.00



	allocation for research and coding	
Technical Assets	Premium 2D UI Kits, Sprite Sheets, and Sound FX Licenses	₱3,500.00
Office Supplies	High-yield Ink Cartridges, A4 Paper reams, and draft binders	₱3,000.00
Utilities	Electricity overhead for development workstation usage	₱4,000.00
Subtotal (A)		₱109,000.00
II. Implementation & Maintenance		
Infrastructure	Cloud Hosting for Python Backend and MySQL Database Storage	₱6,200.00
Network & Security	Custom Domain Registration (.com) and SSL Certificate	₱1,300.00
Focus Group Support	Incentives and refreshments for the 20 Focus Group participants	₱2,500.00
Final Documentation	Hard-binding of final thesis copies and evaluator materials	₱5,000.00
Maintenance Reserve	Budget for post-launch bug fixes and system optimization	₱3,500.00
Subtotal (B)		₱18,500.00
GRAND TOTAL (A + B)		₱127,500.00

The researchers calculated the project's total economic valuation at ₱127,500.00. The largest component is the Development Subtotal (₱109,000.00), which primarily reflects the value of the technical labor provided by the proponents. While the labor costs are simulated and do not require actual cash disbursement, the out-of-pocket expenses for items such as printer supplies, high-speed WiFi, and electricity are critical for the completion of the system. The Implementation Subtotal (₱18,500.00) ensures that the platform remains stable and accessible to the 100 respondents during



the evaluation phase. Based on this financial map, the proponents conclude that the project is manageable and financially sound for a student-led research team.

A. Benefits and Return of Investment (ROI)

- **Intangible Benefits:**

- ❖ Provides a stress-free cognitive assessment tool for students.
- ❖ Reduces the cost of professional cognitive profiling for educational institutions.
- ❖ Promotes "Serious Games" as a valid academic and industry-linked research solution.

- **Tangible ROI:**

- ❖ If the system is sold to schools as a yearly subscription (B2B) at ₱10,000 per school, the project reaches a "break-even" point with only 3 institutional clients.
- ❖ Calculation: Total Cost (₱23k) ÷ License Fee (₱10k) = 2.3 Years (Payback Period) with very low volume.

B. Commercialization and Monetization Plan

To ensure the long-term sustainability of CogniCore, the following strategies will be implemented:

1. **B2B (Business-to-Business) Licensing:** Selling the platform to schools and HR departments as a tool for student/employee cognitive screening.
2. **Freemium Model (B2C):**
 - **Free:** Users can play the four mini-games and see basic scores.
 - **Premium:** Users pay a small one-time fee (₱99.00) to unlock the full AI-Driven Analytics Report and download their 360-degree radar charts as a PDF.



3. **Data Partnership:** Anonymized data trends can be provided to educational researchers to help improve student learning strategies (adhering to Data Privacy laws).

2.7.1.3 - Development and Operational Cost

In order to make sure that CogniCore not only works in terms of technology but is also sustainable, we designed a market strategy. The strategy highlights who our audience is and how the platform can be scaled past the capstone requirement.

Table 5: Market Plan

Category	Team Strategy
Target Market	Our team has identified Higher Education Institutions (HEIs) and Corporate HR Departments as our primary market. These organizations require reliable, non-clinical screening tools to assess the mental agility and cognitive focus of students and prospective employees.
Competitive Edge	Unlike traditional static assessment tools, the researchers implemented AI-driven Dynamic Difficulty Adjustment (DDA). This allows CogniCore to eliminate "test anxiety," providing a more natural and accurate cognitive profile compared to standard numerical tests.
Distribution Strategy	The platform will be distributed as a Software-as-a-Service (SaaS) model. Because it is web-based, it can be accessed instantly via desktop or mobile browsers without requiring the user to download heavy software.



Promotion Plan	The team plans to promote CogniCore through academic partnerships and research symposiums. By demonstrating the system's effectiveness during our Pilot Run at the Molino Campus, we can establish a local success story to attract further institutional interest.
Sustainability	To maintain the system, we will adopt a "Freemium" model. The four core cognitive games will remain free for general users, while advanced AI-generated PDF Analytics Reports and long-term historical tracking will be available via a premium subscription.

The design of this market plan by our group was made with the intention of linking theoretical knowledge gained through research to its practical application. In the creation of this plan, emphasis has been laid on using the "Freemium" sustainability model, which will ensure that the platform continues to be open for use by any students as well as generating enough revenue for keeping alive the servers used.

2.7.2 - Software Design

2.7.2.1 - Conceptual Design

The conceptual design of CogniCore follows the **Input-Process-Output (IPO) model**. It focuses on the high-level transition of player behavior into a cognitive profile.

- **Input:** User credentials and real-time gameplay data (clicks, reaction time, errors).
- **Process:** The system captures metrics via Phaser.io, adjusts difficulty through DDA logic, and classifies the data using a Random Forest AI model.

- **Output:** A visual Analytics Dashboard featuring a 360-degree Radar Chart and a specific "Cognitive Archetype" profile.

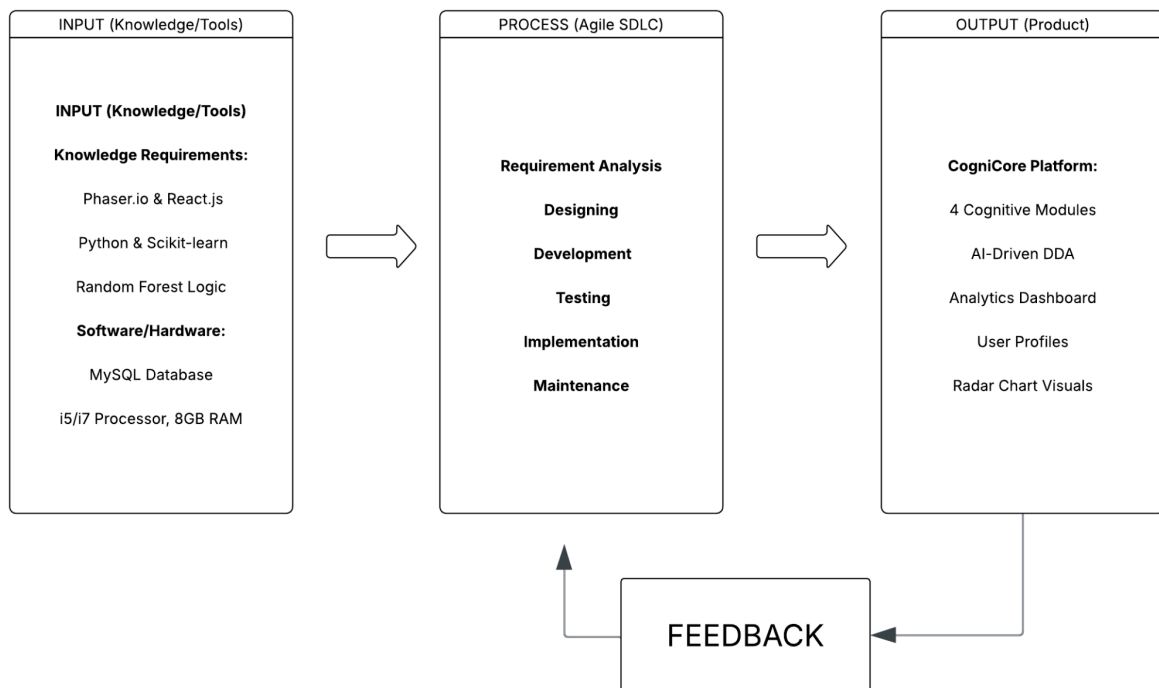


Figure 3: Conceptual Model of the Study

The figure above illustrates the conceptual structure that guides the researchers. The Input phase consists of the technical knowledge and hardware needed. The Process follows the standard Agile stages to develop the code and AI logic. The Output is the final functional platform providing cognitive insights and performance analytics.

2.7.2.2 Technical Architecture and Data Flow

In order to create a proper technical structure for CogniCore, we chose to use Gane-Sarson Notation for all Data Flow Diagrams (DFD). We decided to use Gane-Sarson notation specifically since it is considered to be an industrial standard for software engineers. As opposed to a simple logic flow chart, Gane-Sarson notation uses a rounded rectangle for processes, square for external entities, and an open rectangle for data storage. This will enable the researchers to differentiate between data flows and application logic within the system.

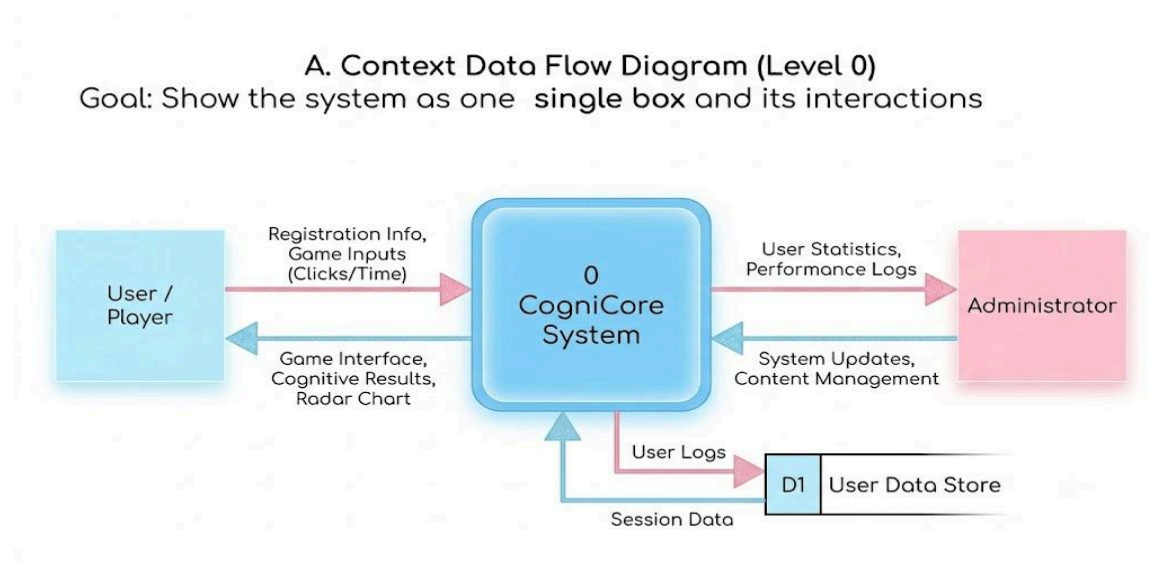


Figure 4: Context Data Flow Diagram

In designing the Context DFD, we aimed to establish the scope of CogniCore as a whole. This DFD shows how the core application is the hub connecting the two main components: the User/Player and the Administrator. It is shown how the User gives the input from registration data and real-time gameplay, and in return, receives dynamic game interface and

cognitive feedback from the artificial intelligence. At the same time, the Administrator sends the input of system updates and retrieves performance logs from the system.

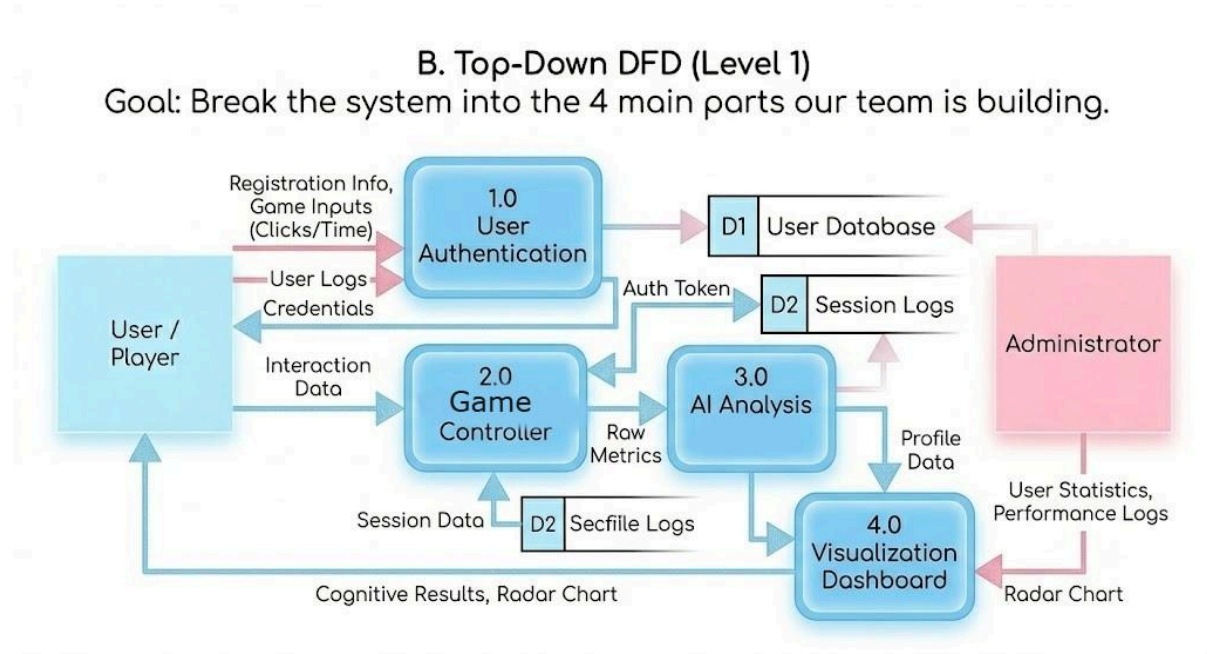


Figure 5: Top-Down Data Flow Diagram

In this Top Down DFD, the authors have further divided the system into four different functional processes: User Authentication (1.0), Game Engine Controller (2.0), AI Analysis Engine (3.0), and Visualization Dashboard (4.0). The above diagram is essential for us as it clearly defines how an internal store like the User Database (D1) and Session Logs (D2) is used. It helps us understand how an interaction from the game engine is converted into a numerical value and recorded.

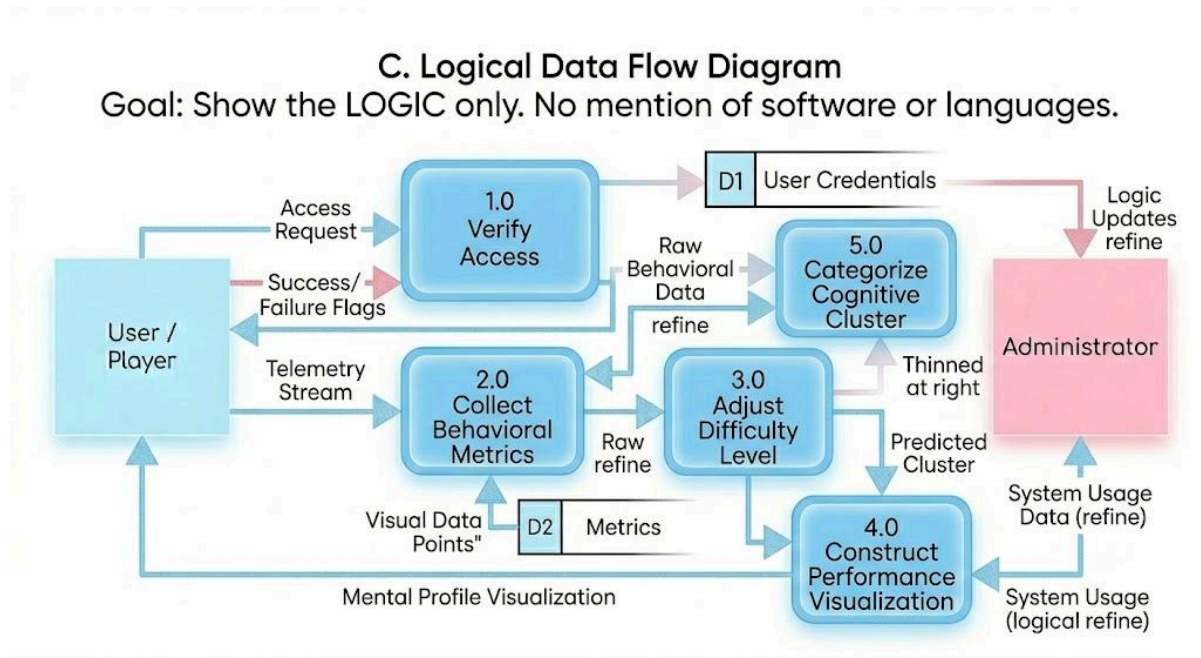


Figure 6: Logical Data Flow Diagram

The Logical DFD strictly concerns itself with the logic behind CogniCore, irrespective of what programming languages are used. The logic behind the design is what our team concentrated on while designing our Logical DFD, which includes identifying the logic behind validating identity, behavior collection, modifying difficulty level dynamically, and classifying people according to their cognitive levels. Thus, by concentrating on the logical aspect of design, we ensured that the "Mental Profile Visualization" is based upon data analysis in a systematic way.

D. Physical Data Flow Diagram

Goal: Show the TECH. This proves we are BSIT students.

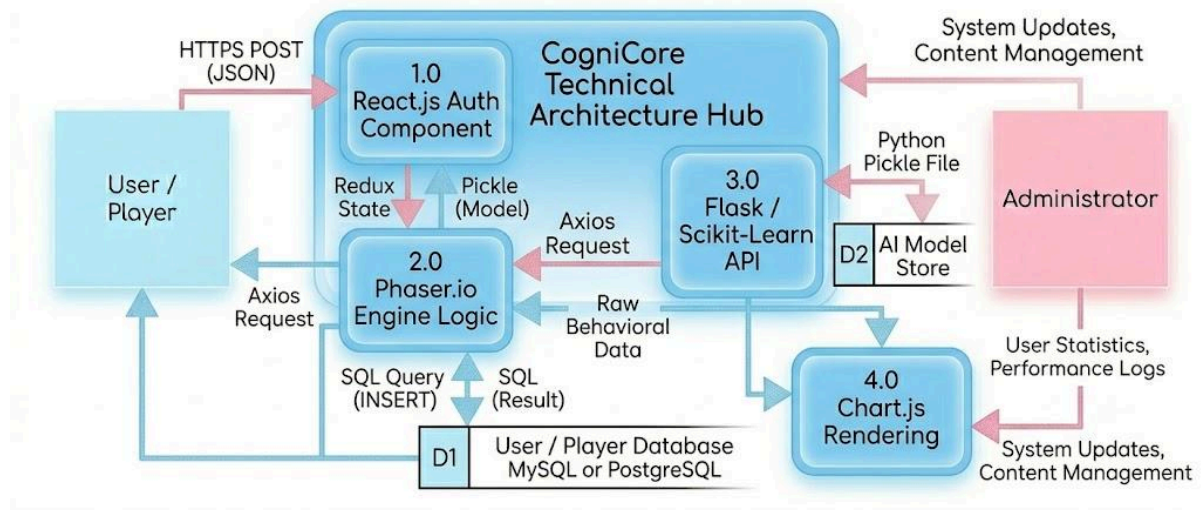


Figure 7:Physical Data Flow Diagram

In order to have a more clear approach towards implementation, we created the Physical DFD which outlines the exact technology stack that our team will be using, including React.js as the front-end, Phaser.io as the engine, and Python and Flask for the AI. We have also defined the exact technical specifications which include HTTPs POST request for JSON data transfer and SQL queries for database management. Through this, we can ensure that all members of our team are on the same page regarding the integration between our JavaScript game and Python model.

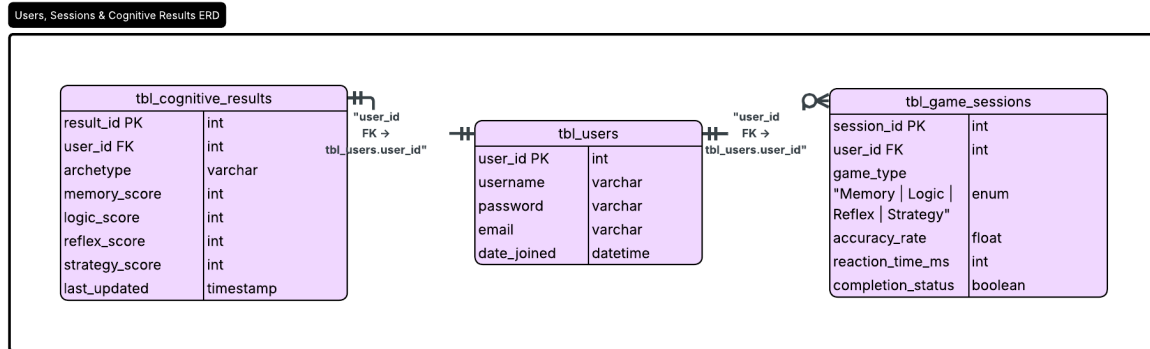


Figure 8: Entity Relationship Diagram

The Entity Relationship Diagram (ERD) above defines the database structure for CogniCore. It maps the relationships between the user accounts, the stored gameplay telemetry, and the final AI-generated profiles. The researchers ensured that the data architecture supports "One-to-Many" relationships between users and their game sessions to track progress over time.

A. Data Dictionary

Table 6: *tbl_users*

Field Name	Data Type	Size	Constraint	Description
user_id	INT	11	PK, AI	Unique identifier for each user (Primary Key).
username	VARCHAR	50	Unique, NN	The unique name chosen by the user for login



password	VARCHAR	255	NN	The hashed security credential for the account.
email	VARCHAR	100	Unique, NN	User's email for verification and recovery.
date_joined	TIMESTAMP	-	Default: Now	Automatic timestamp of when the account was created.

tbl_users presents the data dictionary for the users table used in the CogniCore system. The table defines the fields, data types, constraints, and descriptions of the information stored for each registered user. It includes essential account details such as the user ID, username, password, email address, and date created. This table serves as the primary repository for user account management, authentication, and identification, ensuring secure access to the system and proper association of cognitive assessment records with individual users.

Table 7: tbl_game_modules

Field Name	Data Type	Size	Constraint	Description
module_id	INT	11	PK, AI	Unique identifier for the game module.
module_name	VARCHAR	20	NN	Name of domain (Memory, Logic, Reflex,



				Strategy).
difficulty_cap	INT	3	NN	The maximum difficulty level reachable in this module.
description	TEXT	=	-	A brief explanation of what the mini-game measures.

tbl_game_modules presents the data dictionary for the game modules table used in the CogniCore system. The table defines the fields, data types, constraints, and descriptions of the information stored for each cognitive game module. It includes the module ID, module name, difficulty cap, and module description. This table serves as the reference for all cognitive game categories in the system, such as Memory, Logic, Reflex, and Strategy, and helps organize the modules used for cognitive assessment and training.

Table 8: tbl_game_sessions

Field Name	Data Type	Size	Constraint	Description
session_id	INT	11	PK, AI	Unique identifier for each specific game session.



user_id	INT	11	FK	Reference to the player (Foreign Key).
module_id	INT	11	FK	Reference to the game type played (Foreign Key).
raw_score	INT	11	NN	The total points earned during the session.
accuracy_rate	DECIMAL	5, 2	NN	Percentage of correct moves (e.g., 0.85 for 85%).
reaction_ms	INT	11	NN	The average response time measured in milliseconds.
session_date	DATETIME	-	NN	The exact date and time the session was completed.



tbl_game_sessions presents the data dictionary for the game sessions table used in the CogniCore system. The table defines the fields, data types, constraints, and descriptions of the data recorded for every game session. It stores information such as the session ID, user ID, game module played, raw score, accuracy rate, reaction time, and session completion date. This table enables the system to track player performance, record gameplay statistics, and provide the necessary data for AI-driven cognitive assessment and analytics.

Table 9: tbl_cognitive_profiles

Field Name	Data Type	Size	Constraint	Description
profile_id	INT	11	PK, AI	Unique identifier for the AI-generated report.
user_id	INT	11	FK	Reference to the owner of the profile (Foreign Key).
archetype	VARCHAR	50	NN	The AI classification (e.g., "The Speedster")
memory_rating	INT	3	NN	Scaled score (1-100) for Memory



				performance.
logic_rating	INT	3	NN	Scaled score (1-100) for Logic performance.
reflex_rating	INT	3	NN	Scaled score (1-100) for Reflex performance.
strategy_rating	INT	3	NN	Scaled score (1-100) for Strategy performance.
last_updated	TIMESTAMP	-	NN	When the AI last recalculated this user's profile.

Legend:

PK: Primary Key

FK: Foreign Key

AI: Auto-Increment

NN: Not Null (Required field)



tbl_cognitive_profiles presents the data dictionary for the cognitive profiles table used in the CogniCore system. The table defines the fields, data types, constraints, and descriptions of the data stored for each user's cognitive assessment. It includes information such as the user's unique profile ID, associated user ID, cognitive archetype classification, memory score, logic score, reflex score, strategy score, and the date of the latest profile update. This table serves as the central repository for storing and managing users' cognitive performance results and AI-generated cognitive profiles.

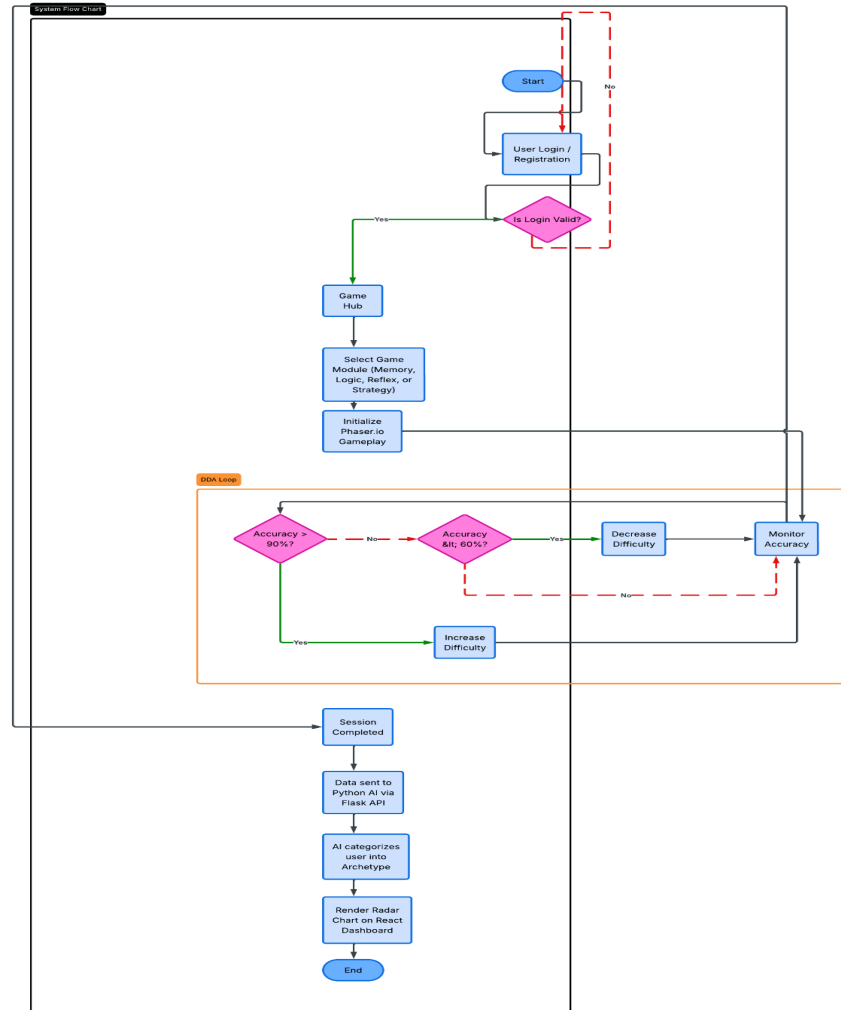


Figure 9: System Flow Chart

The System Flowchart describes the logical path a user takes from login to the final result. It highlights the decision points within the DDA loop, where the system checks the player's accuracy to either increase or decrease difficulty, ensuring a seamless transition from gameplay to data processing.



H. Algorithm Structure

CogniCore utilizes two primary algorithms to function:

A. Dynamic Difficulty Adjustment (DDA):

- *Input:* Player's current success rate and time-on-task.
- *Logic:* A conditional loop that checks performance every 10 seconds.
- *Output:* Real-time updates to game speed, number of enemies, or complexity of puzzles.

B. Random Forest Classification (AI Model):

- *Input:* 4 Features (Accuracy, Reaction Time, Error Frequency, Completion Speed).
- *Logic:* An ensemble learning method that uses multiple decision trees to predict which "Cognitive Archetype" best fits the player's behavioral data.
- *Output:* A classification label (e.g., "The Analytical Thinker").

A. AI Tools and APIs

- **Scikit-Learn:** The Python library used to train and deploy the Random Forest model.
- **Flask API:** The bridge used to transmit gameplay JSON data from the JavaScript-based game to the Python AI server.
- **Chart.js API:** Used in the React frontend to transform numeric AI data into a visual Radar Chart.

B. Story Board (for Games only)



1. **Scene 1 (The Welcome Screen):** Minimalist UI with "Start Assessment" and "How to Play" buttons.
2. **Scene 2 (The Game Hub):** A map or grid showing the four cognitive domains (Memory, Logic, Reflex, Strategy).
3. **Scene 3 (Active Gameplay):** Example: Memory Game – A 4x4 grid where tiles flash in sequence, and the player must repeat the pattern.
4. **Scene 4 (The Analytics Lab):** The final screen showing the user's Radar Chart and a text description of their Cognitive Archetype.

C. Mechanics (for Games only)

- **Core Loop:** Observe Challenge → Interactive Response → Receive Feedback/DDA adjustment.
- **Scoring:** Points are awarded based on speed and accuracy. Deductions are made for "Miss-clicks" or "Logic Errors."
- **Win/Loss Condition:** Most games are time-based (e.g., 2 minutes per module). The "Win" is completing the assessment; the "Loss" is failing to complete the sequence within the allotted time.

2. 7. 3 - Testing Procedures

The testing phase ensures that all components of CogniCore—the Phaser.io game modules, the Python AI backend, and the React dashboard—work together seamlessly. The researchers will follow a multi-stage testing process:



- **Alpha Testing (Internal):** Conducted by the developers to identify immediate "game-breaking" bugs, such as crashes during scene transitions or database connection failures.
- **Beta Testing (External):** A controlled test involving a small group of users to check for gameplay balance and user interface clarity before the official Pilot Run.
- **Black Box Testing:** Testing the software without looking at the internal code, focusing purely on inputs (gameplay actions) and outputs (score results and AI labels) to ensure the system behaves as expected.
- **White Box Testing:** Examining the internal logic of the scripts, specifically the Dynamic Difficulty Adjustment (DDA) algorithm and the Random Forest Classifier, to ensure data is being processed correctly.

System Test Plan

The System Test Plan outlines the specific areas of the application that will be evaluated to ensure the project meets its functional requirements.

Table 10: System Test Plan

Test Area	Objective	Methodology	Expected Result
Authentication	To verify that users can securely register and log in.	Inputting valid/invalid credentials and checking session persistence.	Only authorized users can access the game and their personal dashboard.
Game Logic	To ensure the 4 cognitive modules	Playing through each module	Game transitions, scoring, and timers



	(Memory, Logic, etc.) run without errors.	multiple times.	function according to the mechanics design.
DDA Functionality	To verify that the game scales difficulty in real-time.	Intentionally performing well or poorly during a session.	The game speed/complexity increases or decreases based on performance.
AI Integration	To ensure gameplay data is correctly classified by the Python backend.	Sending varying datasets (high accuracy vs. low accuracy) to the AI.	The AI returns a "Cognitive Archetype" that logically matches the input data.
Data Persistence	To check if scores and profiles are saved in MySQL.	Completing a game and checking the database records.	Data is accurately stored and can be retrieved on the User Dashboard.
Responsiveness	To ensure the web-based game works on different screen sizes.	Accessing the URL via Desktop, Laptop, and Mobile browsers.	The UI layout and game canvas scale appropriately to the screen resolution

System Test Plan presents the test areas, objectives, methodologies, and expected results used to evaluate the functionality and performance of the CogniCore system. The table covers critical components such as Authentication, Game Logic, Dynamic Difficulty Adjustment



(DDA), AI Integration, Data Persistence, and Responsiveness. Each test ensures that the system operates correctly, processes user data accurately, adjusts gameplay difficulty appropriately, securely stores information, and remains accessible across different devices and screen sizes. The results of these tests help verify the reliability, functionality, and overall performance of the developed system.

2.7.4 Deployment

Since **CogniCore** is a web-based "Serious Game" platform, the deployment will follow a Cloud-Based Hosting strategy to ensure high accessibility for the respondents and evaluators.

- **Frontend Deployment:** The React.js dashboard and Phaser.io game assets will be deployed using **Vercel or Netlify**. These platforms provide fast loading speeds for static assets and global CDN support.
- **Backend Deployment:** The Python (Flask) API and the Machine Learning model will be hosted on Render or **PythonAnywhere**. These services allow the AI model to remain "always-on" to process incoming gameplay data.
- **Database Hosting:** The MySQL database will be hosted using a cloud database provider (such as **Aiven or Clever Cloud**) to allow real-time data syncing between the frontend and the AI backend.
- **Deployment Pipeline:**
 1. Pushing code updates to **GitHub**.
 2. Automatic triggers for "Build and Deploy" on the cloud hosting platforms.
 3. Final URL verification and SSL (HTTPS) certificate activation to ensure data security for users.



2.8 Testing and Evaluation

2.8.1 - Unit Testing

The researchers will perform unit testing to verify that individual components of the source code function correctly in isolation.

- **Game Module Tests:** Verifying that the scoring logic in each of the 4 Phaser.io games calculates points accurately.
- **DDA Logic Tests:** Testing the script that triggers difficulty changes (e.g., ensuring speed increases exactly when the 90% accuracy threshold is met).
- **API Endpoint Tests:** Using tools like Postman to ensure that the Flask backend correctly receives JSON data and returns the proper AI classification.
- **Database Tests:** Verifying that the "Insert" and "Update" queries for player profiles work without data corruption.

2.8.2 - Usability Testing

To ensure the game is intuitive and engaging, usability testing will be conducted focusing on the User Interface (UI) and User Experience (UX).

- **Learnability:** How quickly can a new player understand the mechanics of the mini-games?
- **Navigability:** Is the transition between the Game Hub and the Analytics Dashboard smooth and logical?
- **Visual Clarity:** Are the Radar Charts and AI reports easy to interpret for a non-technical user?



2.8.3 - System Testing

System testing evaluates the end-to-end integration of all project parts. The researchers will follow a "Full Loop" test:

1. **Start:** User creates an account.
2. **Middle:** User plays 15 minutes of various cognitive modules.
3. **Process:** Data is sent to the Python AI, and the database is updated.
4. **Finish:** User views their updated Radar Chart and Archetype on the React Dashboard.

Success is defined as the system completing this entire loop without manual intervention or errors.

2.8.4 - Pilot Run

The Pilot Run is the official deployment of the system to the **thirty (30) target respondents**.

- **Environment:** The test will be conducted in a controlled setting (e.g., a computer lab or a monitored online session).
- **Procedure:** Participants will be given a brief orientation, play the game for a full session, and then immediately answer the Evaluation Questionnaire.
- **Data Collection:** This run provides the primary dataset for the Statistical Treatment of Data.



2.8.5 - Evaluation rubrics (ISO 25001 Questionnaire)

The system will be evaluated by both users and experts based on the following quality characteristics.

Table 11: Evaluation rubrics (ISO 25001 Questionnaire)

Characteristic	Sub-Characteristic	Description
Functional Suitability	Functional Correctness	Does the system provide accurate cognitive profiling and game results?
Performance Efficiency	Time Behavior	Does the game load quickly and the AI process data without significant lag?
Usability	Appropriateness Recognizability	Is the game's purpose and interface clear to the target student users?
Reliability	Maturity/Availability	Does the system remain stable and accessible throughout the entire play session?
Security	Confidentiality	Are user credentials and cognitive profiles protected from unauthorized access?



Maintainability	Modularity	Is the code organized so that new game modules can be added easily in the future?
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Evaluation Rubrics (ISO 25010 Questionnaire) presents the software quality characteristics and sub-characteristics used to evaluate the CogniCore system based on the ISO 25010 Software Quality Model. The table includes Functional Suitability, Performance Efficiency, Usability, Reliability, Security, and Maintainability, along with their corresponding evaluation criteria. These quality attributes serve as the basis for assessing whether the system meets user requirements, performs efficiently, remains reliable and secure, and can be maintained or enhanced in the future. The results obtained from this evaluation will help determine the overall quality and effectiveness of the developed system.

Evaluation Scale (Example for Questionnaire)

Participants will rate each item using a 5-point Likert Scale:

5 – Excellent / Strongly Agree

4 – Very Good / Agree

3 – Good / Neutral

2 – Fair / Disagree

1 – Poor / Strongly Disagree



2.9 Ethical Consideration

The researchers are committed to maintaining the highest ethical standards throughout the development and evaluation of CogniCore. The study will strictly adhere to the principles of human rights, data protection, and academic integrity.

2.9.1 - Informed Consent

Prior to the Pilot Run, every participant will be provided with an **Informed Consent Form**. This document will clearly explain:

- The purpose of the study and how the game-based assessment works.
- That participation is entirely voluntary.
- That participants have the right to withdraw from the game at any point without any penalty or explanation.
- That the results are for research purposes only and do not constitute a formal medical or psychological diagnosis.

2.9.2 - Data Privacy

In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), the researchers will ensure that all personal information is handled with extreme care.

- Only the necessary data required for the AI analysis (e.g., gameplay metrics) will be collected.
- No sensitive personal information (such as home addresses or private contact details) will be requested within the application.



- Data transmission between the Phaser.io frontend and the Python backend will be secured to prevent unauthorized interception.

2.9.3 - Confidentiality

The researchers will guarantee the anonymity of all respondents to protect their identity and reputation.

- **Data Anonymization:** Participants will be assigned a "User ID" or a pseudonym. Their real names will not be linked to their specific "Cognitive Archetype" or performance scores in any public report or the final documentation.
- **Secure Storage:** All database records in MySQL will be accessible only to the primary researchers and the project adviser.
- **Aggregate Reporting:** Results will be presented in the study as "aggregate data" (e.g., "60% of the participants scored high in the logic module") rather than individual performance reviews.

2.9.4 - Conflict of Interest

The researchers declare that there is no conflict of interest regarding this study.

- **No Financial Gain:** The proponents have no financial ties to any psychological testing firms or AI software companies that could bias the results of the project.
- **Objectivity:** The AI model (Random Forest) is trained on objective gameplay data to ensure that the profiling is based on mathematical evidence rather than the personal bias of the researchers.
- **Academic Purpose:** The primary goal of the project is to fulfill the requirements for the degree of Bachelor of Science in Information Technology and to contribute to the field of Serious Games and AI analytics.



2.10 Research Instruments Used

The researchers will utilize a set of specialized instruments to collect data, validate the system's logic, and evaluate the user experience.

2.10.1 - Content Validation

To ensure that the four mini-games actually measure the intended cognitive domains (Memory, Logic, Reflex, and Strategy), the "Content Validation" process will be performed. The researchers will map each game mechanic to established psychological benchmarks:

- **Memory:** Based on the N-back task or Visual Working Memory tests.
 - **Reflex:** Based on Simple Reaction Time (SRT) and Choice Reaction Time (CRT) metrics.
 - **Logic/Strategy:** Based on Pattern Recognition and Tower of Hanoi logic structures.
- The validity of these mappings will be reviewed and signed off by the Subject Matter Experts.

2.10.2 - Expert Consultation

To ensure that the research instruments are scientifically sound and technically accurate, the researchers will submit the survey questionnaires for formal validation. A panel of five (5) experts will review the instruments to evaluate their relevance, clarity, organization, and alignment with the research objectives. The feedback provided by these professionals will be used to refine the items before the official data collection phase.



IT Experts (3): The three (3) IT Experts will evaluate the questionnaire specifically in terms of technical content validity. Their role is to ensure that the statements regarding software quality—based on the ISO 25010 model—are technically accurate and clearly understood by respondents. They will check if the questions properly measure functional suitability, performance efficiency, and usability within a web-based game environment.

Subject Matter Experts (2): The two (2) Subject Matter Experts, consisting of licensed psychologists or educators, will review the items for thematic relevance. Their focus is to confirm that the Game Experience Questionnaire (GEQ) and the cognitive assessment items are appropriate for measuring mental agility, "Flow State," and behavioral patterns. Their expertise will strengthen the psychological validity of the instrument, ensuring it correctly captures the intended cognitive metrics.

The researchers will utilize the Content Validity Ratio (CVR) or a standardized validation form to quantify the experts' feedback. Only items that receive a high consensus for relevance and clarity will be included in the final version of the survey, ensuring the reliability of the data gathered from the 100 respondents and the focus group.

2.10.3 - Survey Questionnaire

The primary instrument for user evaluation is a structured Survey Questionnaire based on the ISO 25010 Software Quality Model. The questionnaire will use a 5-point Likert Scale and is divided into the following criteria:

- **Functional Suitability:** Does the AI correctly identify the player's strengths?
- **Usability:** Is the game interface intuitive and the dashboard easy to read?
- **Performance Efficiency:** Does the game run smoothly without lagging?



- **Operability:** Is the "Serious Game" engaging enough to complete the assessment?

2.10.4 - Pre-Test/ Post-Test

This instrument will be administered to the Focus Group of 20 participants. The Pre-test will establish a baseline of the user's cognitive awareness and skill level. After engaging with the CogniCore modules, the Post-test will be conducted to measure any significant changes in performance or cognitive awareness, providing the researchers with empirical data on the system's training effectiveness.

Table 12: Summary of Research Instruments

Instrument	Target Group	Goal
Validation Rubric	IT & Subject Matter Experts	To ensure the system is technically and scientifically valid.
ISO 25010 Survey	30 Student Respondents	To measure user satisfaction and software quality.
Pre/Post Test	30 Student Respondents	To measure the accuracy of the AI assessment vs. manual benchmarks.

Summary of Research Instruments presents the instruments used in the study, their target respondents, and their corresponding purposes. The **Validation Rubric** will be administered to IT and subject matter experts to evaluate the technical and scientific validity of the system.



The **ISO 25010 Survey** will be answered by 30 student respondents to assess software quality and user satisfaction. Meanwhile, the **Pre-Test and Post-Test** will be conducted among the same 30 student respondents to measure the effectiveness and accuracy of the AI-driven assessment system by comparing results before and after system use. These instruments collectively ensure the validity, quality, and effectiveness of the developed system.

2.11 Statistical Treatment of Data

To ensure the validity and reliability of the gathered data, the researchers will apply the following statistical designs and methods. The data will be processed using tools such as Microsoft Excel or SPSS.

2.11.1 - Statistical Design

The study will utilize a **Descriptive and Inferential Statistical Design**:

- **Descriptive Design:** This will be used to summarize the demographic profile of the respondents and the mean scores from the ISO 25010 evaluation.
- **Inferential Design:** This will be used specifically for the Pre-test/Post-test analysis to determine if there is a significant difference in the users' cognitive awareness or performance after interacting with the system.

2.11.2 - Statistical Methods

To systematically evaluate user telemetry records and mathematically analyze software quality validation scores, the gathered dataset will be processed using descriptive and inferential statistical methods. The calculation routines are embedded directly into the platform's backend analytics engine (server/app.py), running programmatically over the



persisted SQLite transaction records via Python data libraries (SciPy and NumPy). The following statistical treatments will be used:

A. Frequency and Percentage Distribution

This descriptive method is utilized to group, categorize, and present the demographic composition of the sample population, as well as the total volume of distinct gameplay tracking runs performed across the 4 core parent domains (Spatial-Visual Memory, Logical-Mathematical Reasoning, Reflexes and Focus, and Executive Strategy).

The percentage formula is expressed as:

$$\text{Percentage} = (f / N) * 100$$

Where:

- Percentage = Percentage distribution score
- f = Frequency of a specific subgroup or logged game session entry
- N = Total sample size or cumulative data pool count of the study

B. Weighted Mean

The weighted mean is implemented to calculate the baseline and aggregate evaluation results obtained from the expert validators and end-user testing groups. These target scores are captured through an interactive frontend Likert-Scale questionnaire built inside the Clinical Researcher View, mapping across the five core quality characteristics of the ISO 25010 Software Quality Model: Functionality, Usability, Reliability, Performance Efficiency, and User Experience.

The weighted mean formula is defined as:

$$\text{Weighted Mean} = \text{Sum}(f * x) / N$$

Where:

- Weighted Mean = Calculated weighted mean score for a given quality metric



- f = Frequency of respondents choosing a particular scale rating entry
- x = Weight assigned to the individual Likert interval level (ranging from 1 to 5)
- N = Total number of evaluating technical and user respondents

The computed numerical means will be formally interpreted using the following institutional boundaries defined by the college:

- 4.51 - 5.00: Strongly Agree (Excellent / Highly Functional)
- 3.51 - 4.50: Agree (Very Satisfactory / Functional)
- 2.51 - 3.50: Neutral (Satisfactory / Moderately Functional)
- 1.51 - 2.50: Disagree (Fair / Minimally Functional)
- 1.00 - 1.50: Strongly Disagree (Poor / Not Functional)

C. Standard Deviation

Standard deviation will be calculated alongside the weighted mean to evaluate the variability, consistency, and consensus of the system validation data provided by the expert evaluators. It determines how tightly the Likert scores cluster around the calculated mean performance boundaries.

The standard deviation formula is expressed as:

$$\text{Standard Deviation} = \text{SquareRoot}(\text{Sum}(x - \text{Weighted Mean})^2 / (n - 1))$$

Where:

- Standard Deviation = Calculated sample standard deviation score
- x = Individual Likert response rating entry score value
- Weighted Mean = Computed weighted mean score for that specific target metric section
- n = Total number of completed expert validation instruments received ($n = 30$)

D. Paired Student's t-Test



To evaluate the overall impact and empirical training effectiveness of CogniCore, an inferential statistical analysis will be run on a 30-participant testing cohort. The platform's automated evaluation route (/api/evaluate) uses the Python `scipy.stats.ttest_rel` method to compute differences between paired sample columns.

It checks for a statistically significant reduction in user processing latencies (`reaction_time`, `hesitation_ms`) and physical mechanical errors (`error_count`, `spam_click_count`) tracked across chronological milestone runs (Pre-test baseline metrics vs. Post-test completion metrics).

The paired t-test formula is defined as:

$$t = \text{Mean_of_Differences} / (\text{Standard_Deviation_of_Differences} / \text{SquareRoot}(n))$$

Where:

- `t` = Calculated t-statistic outcome checked against the critical distribution table limits
- `Mean_of_Differences` = Mean of the delta differences between individual participant Pre-test and Post-test values
- `Standard_Deviation_of_Differences` = Standard deviation of the discovered mean delta difference values
- `n` = Total number of paired participant records mapped within the experimental test pool (`n` = 30)

A significance marker parameter of $\text{Alpha} = 0.05$ is enforced. If the programmatic output layer yields a probability metric value of $p < 0.05$, the null hypothesis is rejected, providing mathematical proof that the game-based DDA training loop improves cognitive performance.

E. AI Classifier Accuracy Score

To validate the diagnostic precision of the machine learning layer running on the Flask backend, prediction validation is monitored continuously. The profile matrix calculates the



system's real-time accuracy in grouping users into distinct cognitive archetypes (Beginner, Intermediate, Advanced) based on moving averages of behavioral feature inputs.

The classifier accuracy calculation loop is defined as:

$$\text{Classifier Accuracy (\%)} = (\text{Sum_of_Pc} / \text{Sum_of_Pt}) * 100$$

Where:

- Sum_of_Pc = Cumulative count of correct cognitive profile archetypes predicted by the Random Forest model
- Sum_of_Pt = Total number of matching evaluation cycles processed across historical validation benchmarks

This metric monitors the reliability of the underlying random forest model, ensuring that prediction layers maintain consistency above an established system baseline target of 85% accuracy.

Table 13: . Likert Scale Interpretation Table

Mean Range	Verbal Interpretation
4.51 – 5.00	Strongly Agree (Excellent / Highly Functional)
3.51 – 4.50	Agree (Very Satisfactory / Functional)
2.51 – 3.50	Neutral (Satisfactory / Moderately Functional)
1.51 – 2.50	Disagree (Fair / Minimally Functional)
1.00 – 1.50	Strongly Disagree (Poor / Not Functional)

Likert Scale Interpretation Table presents the rating scale used to interpret respondents' evaluation scores. The table shows the corresponding verbal interpretation for each mean range, where higher mean scores indicate a more positive assessment of the system. A mean



score of **4.51–5.00** is interpreted as **Strongly Agree (Excellent/Highly Functional)**, while a mean score of **1.00–1.50** is interpreted as **Strongly Disagree (Poor/Not Functional)**. This scale helps determine the overall level of user satisfaction, acceptance, and functionality of the system being evaluated.

4. Paired T-Test

This inferential method is embedded natively within the server's research calculation node endpoint (/api/evaluate) using the SciPy statistical library to measure clinical or performance training outcomes. The routine tests for a significant difference between paired chronological performance sets (Pretest arrays vs. Posttest records) captured across a 30-participant cohort sample database footprint. The algorithm processes differences in initial task hesitation frames (hesitation_ms) and mechanical panic indices (spam_click_count) to verify system efficacy programmatically, completely eliminating human evaluation bias or manual analytical scripting errors.

5. AI Accuracy Formula

To evaluate the effectiveness of the Random Forest model in classifying cognitive archetypes, the researchers will use the following Accuracy formula:

Formula:

Accuracy = (Number of Correct Predictions / Total Number of Predictions) x 100